

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 14)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

1

2

0 1 . 2

Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 1 . 3

Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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5

0 2 . 1

Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

1

2

3

0 2 . 2

Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

- 0 2 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

- 0 2 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

10

- 0 3 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

03 . 1

Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]**Answer**

million years

03 . 2

Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

03 . 3

Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

03 . 4

Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

0 3 . 5

Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

1

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2

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10

0 4

The Isle of Eigg (area 31 km², population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 4 . 1

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 4 . 2

Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

1

2

3

0 4 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities. **[2 marks]**

0 4 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy. **[2 marks]**

10

0 5 . 1 State the Brundtland Commission (1987) definition of sustainable development. **[1 mark]**

0 5 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one. **[3 marks]**

0 5 . 3

Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

0 5 . 4

Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

10

06 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

- A Thermal pollution of a river by a power station
- B Flooding a valley to create a reservoir
- C Lowering the water table by groundwater abstraction
- D Acid mine drainage entering a stream

☐
☐
☐
☐

06 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

1

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2

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5

07

The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

07 . 1

Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

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07 . 2

Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

07 . 3

Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

10

08 . 1

Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

08 . 2

Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

08 . 3

Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

10

0 9

Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 9**1**

Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

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0 9**2**

Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

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0 9**3**

Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

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09 . 4

Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

10 . 1

Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

10 . 2

Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

10 . 3

Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

1	0	.	4
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Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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[illegible]

Do not write outside the box

END OF QUESTIONS